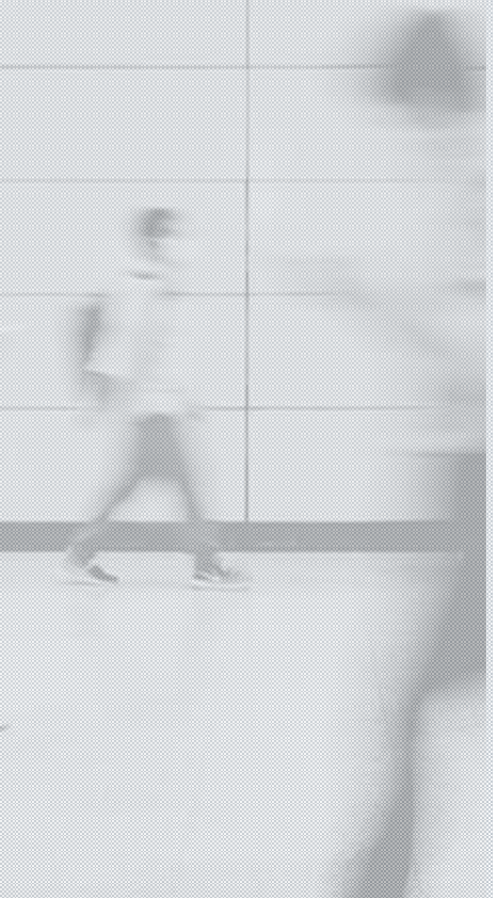
# TREE COUNT & BATCH SIZE



### *More trees: time vs speedup*

Batch 1. Lower latency is better.

| Trees | ONNX ms | Speedup |
|---|---|---|
| 10 | 0.0111 | 54.50× |
| 500 | 0.0200 | 1,051.97× |

### *Best tested batch for throughput*

Among batches 1, 16, 64 and 256

| Trees | Best batch |
|---|---|
| 10 | 256 |
| 100 | 64 |
| 500 | 16 |

### *How to read the speedup*

Each ratio compares the same model with scikit-learn.
500 trees took longer in ONNX, despite the larger speedup.

### *Batching and user waiting time*

Latency measures time per batch.
Throughput measures rows per second.
Waiting to fill a batch is outside this warm CPU benchmark.

AMD tree study, ZipMap off. Source: Final English Report, pp. 8-9.